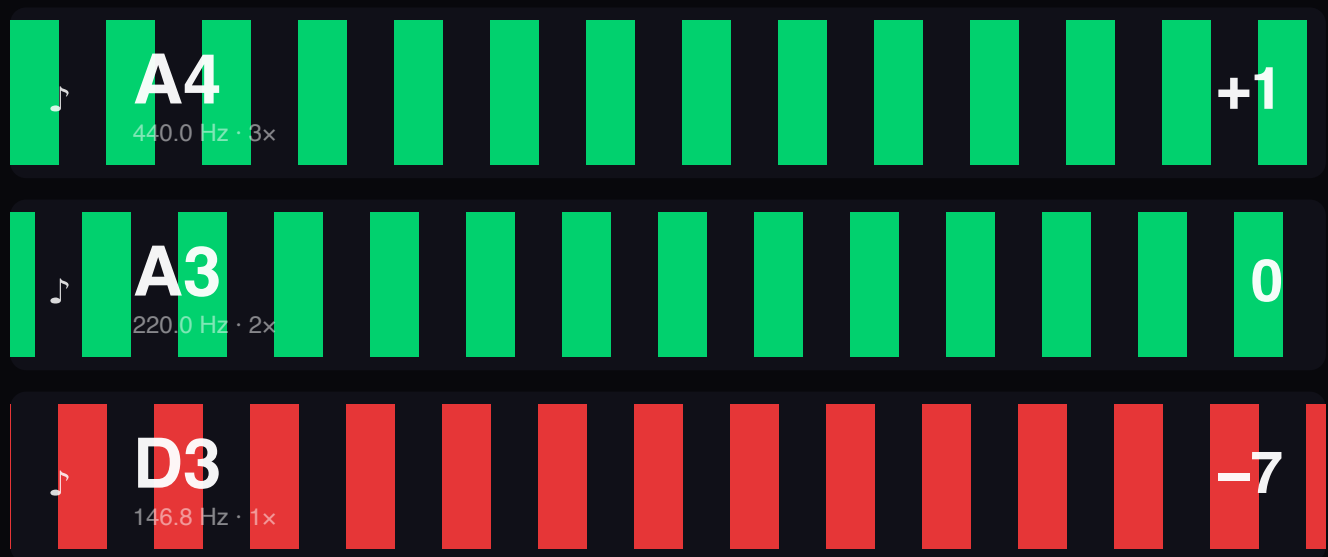


V-Tune

User Guide

A precision strobe tuner for handpans
and other multi-modal instruments.



The three-band strobe display — still + green when locked, drifting + red when out of tune.

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1. Welcome

What V-Tune is and why it exists

V-Tune is a precision strobe tuner built for handpans and other multi-modal instruments — drums, gongs, bells, anything where a single strike excites several pitches at once. Most chromatic tuners give up on these instruments because they're built around one fundamental. V-Tune is built around three: a fundamental, an octave, and an octave-plus-fifth (the 12th), all rendered as independent strobe bands so you can read every partial without them fighting each other.

Under the hood, V-Tune uses FFT peak detection plus phase-rate Goertzel analysis — the FFT tells you which frequencies are loud, the Goertzel measures how stable each one is, frame to frame. You see the result as classic strobe-pattern bars: motion = drift, still = locked.

This guide walks through every part of the app. It's short — V-Tune is a small tool, and the parts that exist exist for a reason. Read it once, then keep tuning.

2. Quick start

From cold app to tuned note in about a minute

1. Grant microphone access

On first launch V-Tune asks for the mic. Allow it. You can change device later in Settings → Input (open Settings with the gear in the teal icon bar).

2. Pick a note

The menu loads open. In the **Tuning / Scale** controls pick a scale from the **SCALE** dropdown and tap a note, or stay on Chromatic and use **OCT -** / **OCT +** to set the octave. The three foundation strobe bands (1×, 2×, 3×) update instantly.

3. Press **Let's Go**

The green button lives at the bottom of the menu (or the bottom of the mobile slide-up). Audio starts. Strike your instrument. The bars freeze when you're in tune; they drift left when you're flat, right when you're sharp.

4. Read the cents number on the right of each band

It reads 0 (and the band washes green) when you're in tune, within $\pm 5\text{c}$ by default. The signed number is exactly how many cents off you are.

Sample strobe band — in tune



3. The menu & utility bar

Where every control lives — and how it gets out of your way

All of V-Tune's controls live in one place: the menu. It looks different on a wide screen versus a phone, but the contents are the same — a teal utility bar up top, the tuning / scale controls, and **Let's Go** pinned to the bottom.

On desktop & landscape tablet

A full-height panel slides out from the right edge. Open and close it with the burger button (three lines in a rounded square) at the top-right of the header. The menu always loads open on launch so everything is to hand.

To give the strobe more room while you tune, the menu auto-hides after 20 seconds of no interaction with it. Any tap, drag or scroll inside the menu resets the timer. If you want it to stay put, hit the pin — that disables auto-hide until you unpin.

If the menu hides while a stopwatch session is running, a compact stopwatch readout appears in the header next to the burger, so your timing stays visible. Tap it to reopen the menu.

On phone & portrait tablet

There's no burger or side panel. Instead the controls live in a bottom quick-pick panel that slides up. Collapsed, it's a single bar with a soft **purple glow** showing the currently-selected note (e.g. "D3"), centred. Tap it to slide the panel up — this pushes the canvas up, which shrinks responsively to make room. It also auto-hides after 20 seconds of no interaction, and has its own pin to keep it open. Full details in section 7.

The teal utility bar

A slim teal-tinted row of icon buttons sits at the top of the menu (and inside the mobile slide-up). The left group, in order:

- Settings (gear) — opens the Settings modal (section 10).
- Stopwatch (stopwatch face) — reveals or hides the stopwatch (section 11).
- Spectrum Analyser (equaliser-bars icon) — reveals or hides the Spectrum Analyser (section 8).

On the right sits the light / dark theme toggle (sun / moon). On mobile the pin icon also lives on the right, just left of the theme toggle. Any active toggle — stopwatch on, spectrum on, pinned — gets a teal highlight so you can see its state at a glance.

4. The strobe display

Where you actually read the tune

The strobe display is the centre of V-Tune. Each horizontal stripe is one band — a frequency the app is tracking. By default you get three foundation bands stacked top-to-bottom:

The display is theme-aware. At rest — when the mic is quiet — the background is a light grey in light mode, or a soft charcoal (not pure black) in dark mode. The moment the mic picks up signal, the background darkens toward a deep near-black for maximum bar contrast, then eases back to its resting tone when you stop. The neutral text (note names, cents) flips light or dark to stay legible against whichever background is showing.

- 3x — the 12th (octave + fifth). Top of the stack.
- 2x — the octave.
- 1x — the fundamental. Bottom of the stack.

A thicker cyan separator marks the boundary between foundation bands and any custom bands you add (via the Spectrum Analyser).

Reading a band

Each band shows you five things at a glance:

- ♪ icon (far left) — the per-band pitch pipe. See section 5.
- Note label — e.g. “D3”. The note this band is tuned to.
- Frequency — the exact target in Hz, under the note label.
- Strobe bars — the moving red/green pattern. Motion direction tells you sharp vs flat.
- Cents readout (far right) — the deviation in cents, signed. Reads 0 (and the band washes green) when in tune.

Colours, and what they mean

 In tune (within $\pm 5\phi$)	 Out of tune
 Selected band	 ♪ Pipe — continuous tone
 ♪ Pipe — beep mode / detected note	 Scale "ding" highlight

Motion = drift

Bars moving left → your input is flat (pitch too low). Tune up. Bars moving right → your input is sharp (pitch too high). Tune down. Bars holding still → you're locked. When a band is in tune it gets a **dark-green wash** and its cents readout settles on 0.

The bars are green in tune and red out of tune, and their blur tracks how far off you are: crisp and sharp when locked, blurring more the further out of tune you drift (and during the unstable attack transient of a fresh strike). Settings → Blur sets the ceiling on that softness.

Peak hold + decay

V-Tune holds the strobe pattern for ~4 seconds after each strike, then fades it out over a second. This means you can let the note die away and still read the tune from the held pattern — no need to keep re-striking. A louder hit resets the hold timer.

5. The pitch pipe (♪)

A reference tone for every band, three clicks deep

Each strobe band has a ♪ icon on its left edge. Click it to hear the band's exact target frequency. Click again to cycle modes:

♪	Off	Silent. The default state.
♪	Tone	Continuous sine reference. Plays the band's frequency forever (until you click again). Useful for hum-along tuning.
♪	Beep	Silent until the band detects a strike — then it fires a brief reference beep at the target frequency. So as you tune each strike triggers a comparison tone you can match against. This is the LinoTune-style mode.

Clicking the ♪ icon on a different band moves the pipe to that band — same mode, new frequency. Only one band can be piping at a time. To stop completely, click the active band's ♪ until it cycles back to Off.

6. The tuning & scale controls

Where you tell V-Tune which note to listen for

On desktop the tuning and scale controls sit directly in the menu, below the teal bar. On phones and portrait tablets they split: the scale picker, note grid and **Let's Go** live in the slide-up quick-pick panel (section 7), while Reference A4 and Tolerance move into Settings → Tuning. Either way the picker has two modes:

AUTO

Tap AUTO to let V-Tune auto-detect whichever note you play, rather than pinning it to a note you've chosen. Handy when you're sweeping across an instrument and don't want to keep re-selecting.

Chromatic mode

A piano-keyboard layout — 7 naturals + 5 sharps. Tap any key to tune to that note in the current octave. Use the OCT – / OCT + buttons below to drop or raise the octave. This is the default mode on launch.

Scale mode

Pick a pre-saved handpan scale from the SCALE dropdown above the keyboard — Kurd, Amara, Pygmy, Aavartan, Ashakiran, Aegean, Equinox and Nordlys, twenty-two layouts in all, with the Ayasa Elements scales at the top of the list. The note picker switches from a chromatic keyboard to just the notes in that scale, each with a fixed octave matching the physical instrument. The **ding** (root) is highlighted in purple. Switching scales auto-selects the ding.

Note naming: in scale mode the accidentals follow the scale's convention (Kurd uses flats, Amara uses sharps). In chromatic mode they follow the ♯ ♭ Do DE buttons in the menu (section 12).

Bottom notes

Notes tuned into the underside of the shell sit in the picker at their true pitch position — often below the ding — and carry a **teal outline** so you can tell them apart at a glance. They tune exactly like any other note; the outline is just there so you know where you are on the instrument.

Gu port notes

Some instruments carry one or two notes tuned into the Gu opening underneath. Where a scale has them, they appear as an **amber chip** beneath the note grid — D Kurd 13 / 15 show E5 · A5, E Amara 13 / 15 show B₅ · D6. They aren't playable targets, so you can't select one the way you select a note.

Tap the chip and V-Tune aims itself at them: both Spectrum Analyser isolation windows jump to ± 35 cents around the notes, the analyser zooms in to frame them, and the two strobe bands underneath then read nothing but the Gu port. Each note takes the colour of the band reading it — **teal** for the lower, **purple** for the higher — so there's no guessing which is which. Tap it again to hand the windows back to their defaults. Full detail on those windows in section 8.

Live note indicator

While audio is running, the note V-Tune is currently picking up from the microphone gets a soft cyan glow ring around its button. This is purely informational — it doesn't change which note you're tuning against. Useful for quickly orienting yourself when you strike an unfamiliar partial.

Tuning reference — PURE vs EQUAL

Above the note layout is a PURE / EQUAL toggle that decides what the three foundation bands are measured against:

- PURE (default) references each band against an exact integer multiple of the fundamental — 1×, 2×, 3×. A handpan whose partials are tuned to those pure ratios reads 0 on every band. This is the acoustically correct reference for handpan partial tuning.
- EQUAL references each band against the nearest equal-tempered note instead. On a pure handpan the compound-fifth (3×) band then reads about +2 cents — the real, constant difference between a pure 3:1 fifth and a tempered fifth. Use this if you tune your partials to equal temperament.

The octave (2×) band reads the same either way — octaves are an exact 2:1 in both systems. Only the compound fifth differs.

Reference A4, Tolerance & Let's Go

Reference A4 sets concert pitch (default 440 Hz) and Tolerance sets how close counts as in tune (default ± 5 cents). On desktop these sit in the menu alongside the picker; on mobile they move into Settings → Tuning. **Let's Go** — which starts and stops audio — is pinned to the bottom of the menu on desktop, and the bottom of the slide-up on mobile.

7. Mobile quick-pick panel

The controls, condensed for thumbs

On phones and portrait tablets there's no burger or side panel. The controls live in a bottom quick-pick panel that slides up. Collapsed, it's a single bar with a soft **purple glow** showing the currently-selected note (e.g.

“D3”), centred. Tap the bar to slide the panel up — this pushes the canvas up, which shrinks responsively to make room.

Opened, it carries the same teal utility bar (settings, stopwatch, spectrum, pin, theme) and the scale / note controls:

- SCALE dropdown — same scales as desktop, above the note grid.
- Chromatic / Scale note grid — a responsive grid of notes, each labelled with its octave; OCT $\pm$ in chromatic mode.
- Let's Go — pinned to the bottom of the slide-up.

Like the desktop menu, the slide-up auto-hides after 20 seconds of no interaction so the strobe gets full width, and it has its own pin (in the teal bar) to keep it open. When it re-collapses it just shows the selected note, centred, ready to tap again. Reference A4 and Tolerance live in Settings → Tuning on mobile.

8. Spectrum Analyser & Isolation windows

See the full frequency content, then isolate the bits you care about

Toggle the Spectrum Analyser with its icon (equaliser bars) in the teal utility bar. It appears under the strobes as a frequency-domain view across the audible range — a real-time picture of every harmonic your instrument is producing, with two isolation bands beneath it for fine-tuning partials.

Two names worth keeping straight, because they are two different things: an isolation window is the coloured bracket you place on the spectrum, and an isolation band is the strobe it drives underneath. You move the window; you read the band. They share a colour so it is always obvious which belongs to which, and the app labels both ISO for short.

Turning the Spectrum Analyser on always restores the two default isolation windows — the teal and purple ones — every time, even if you previously cleared them. So it's never revealed empty.

Isolation windows (ISO)

V-Tune starts with two isolation windows ready to go — a teal one and a purple one. Each is a bracket on the spectrum; V-Tune finds the loudest peak inside the bracket and drives a dedicated strobe band from it, shown beneath the spectrum. The band carries the same colour as its bracket, so it's always clear which window feeds which band.

Drag either end of a bracket to move it; the frequency / note / cents readout follows so you can line an edge up against a note. This is how you tune partials that aren't the fundamental, octave or 12th — say, a particular overtone. Remove a window with its $\times$ button, and draw a new one any time with Shift + drag (or touch-hold then drag on mobile) across the spectrum; you can have up to two at once, splitting the band area 50/50. A re-added window reclaims the freed colour slot.

The isolation bands read frequency with the same sub-cent precision as the main strobe — the same phase-rate physics, not a coarser reading off the FFT peak. What you see on an ISO band is as trustworthy as what you see on the 1 $\times$ / 2 $\times$ / 3 $\times$ bands.

Jumping straight to the Gu port

On a scale with Gu port notes you don't have to place the brackets by hand. Tap the amber Gu port chip under the note grid and both windows land on the port notes at ± 35 cents, with the analyser zoomed to frame them — wide enough to watch the peak float inside the bracket, tight enough that nothing else in the shell wanders in. Tap it again to restore the default windows and the full view. See section 6.

9. Keyboard shortcuts

Play the note picker from a computer keyboard

On desktop you can drive the whole picker from the keyboard without reaching for the mouse — useful when one hand is on the instrument. Hit the **MAP** button at the right of the notation row in the menu for an on-screen reminder of everything below.

Note keys

The letter keys are laid out like a piano: the home row is the white keys, the row above is the black keys, sitting where they would on a real keyboard.

A S D F G H J	The naturals — C D E F G A B, in the current octave.
W E T Y U	The accidentals — C# D# F# G# A#, grouped in twos and threes like the black keys.

Everything else

↑ / ↓	Octave up / down.
← / →	Nudge the target by a cent. Hold Shift for ±10.
0	Reset that nudge back to zero.
Q	Toggle AUTO note detection.
+ / −	Reference pitch (A4) up / down.
Space	Slides the main menu in and out on a wide window. On a narrow one, where there is no side menu, it starts / stops the tuner instead.
Enter	Start / stop the tuner — the keyboard’s Let’s Go.
Esc	Deselect the current strobe band.

Shortcuts stay out of the way while you’re typing in a field or using a dropdown, so they never fire by accident.

10. Settings

Every knob, what it does

Settings is a modal, opened by the gear in the teal utility bar. It’s organised into labelled teal-header sections, each laid out in two columns — the setting’s name and a short description on the left, its control on the right.

V-Tune Pro

Status	Where you stand — Free trial with the days left, or Unlocked once you've bought it. The line underneath says which account you're signed in as. Full detail in section 13.
Unlock forever	Shown during the trial: opens the unlock screen so you can buy without waiting for the trial to run out.

Input

Microphone	Which audio input device V-Tune listens to. Default uses the system mic.
Microphone Sensitivity	Input gain in dB (MIC +/-). Bump it up if your input is quiet, drop it if you're clipping.
Hum	Mains-hum notch filter: Off / 50 Hz (UK/EU) / 60 Hz (US). Notches out mains hum plus its harmonics.

Tuning (mobile only)

On phones and portrait tablets, Reference A4 and Tolerance live here. On desktop these controls sit directly in the menu instead, so this section only appears on mobile.

Reference A4	Concert pitch. Default 440 Hz. Set to 442 for some orchestral work, 432 if you're into that.
Tolerance ($\pm$ cents)	How close you have to be before V-Tune calls you in tune. Default is ± 5 cents.

Strobe Preferences

Brightness	How vivid the red/green strobe bars are. Lower for ambient lighting, higher for stage / sunlight.
Blur	Edge softness of the bars — sharp when locked, automatically softer when way out of tune. This sets the ceiling on that softness.
Speed	How fast the strobe reacts: 0.5x / 1x / 2x / 3x / 5x. Higher = livelier, lower = calmer / easier to read.

Accessibility Options

High contrast	Boosts contrast throughout the UI for easier reading.
Larger text	Scales up the interface text.
Onboarding tour	Replays the interactive guided walkthrough from the start (section 12).

11. Stopwatch

Time your tuning sessions

Toggle the stopwatch on and off with its icon in the teal utility bar. When on, its panel appears pinned at the bottom of the menu, just above **Let's Go** (on mobile, just above the slide-up). Start, stop and reset controls sit on the panel. It counts continuously even if you toggle the panel off and back on.

On desktop, if the menu auto-hides while a session is running, a compact stopwatch readout appears in the header next to the burger so your timing stays visible. Tap it to reopen the menu.

12. Theme, notation, and the onboarding tour

A few preferences worth knowing about

Light / dark theme

Flip between light and dark with the sun / moon icon on the right of the teal utility bar. Fresh installs default to light mode; switch to dark whenever you prefer. The strobe display follows the theme too — see section 4.

Notation

Pick how accidentals are labelled with the four small buttons in the menu, on the row below the note picker: # sharps, ♭ flats, Do solfège, or DE German naming (which uses H for B and B for B $\flat$).

They only appear in chromatic mode — a scale brings its own convention with it (section 6).

The onboarding tour

V-Tune ships with an interactive, learn-by-doing tour. It spotlights each part of the UI and waits for you to actually perform the action before moving on — open the menu, open Settings and walk its sections, reveal the stopwatch and the Spectrum Analyser, pin the menu, pick a scale, then **Let's Go**. It runs automatically on first launch.

Want it again? Open Settings and hit Onboarding tour under Accessibility Options — it restarts the whole guided tour from the top, any time.

Staying up to date (desktop)

The macOS, Windows and Linux apps update themselves. When a newer version is published, V-Tune notices on launch and offers a one-click Install & Restart — no need to redownload or reinstall. (iOS and Android update through their stores; the web app at app.vtune-app.com refreshes itself.)

13. V-Tune Pro

The trial, the one-time unlock, your account

Every install starts with a 14-day free trial of the complete app — no card, no sign-up, nothing withheld. When it ends, a single £49.99 purchase unlocks V-Tune forever. It is not a subscription and there is nothing to renew.

One purchase, every device

The unlock belongs to your account, not to a machine. Buy once and sign in on the Mac in the workshop, the iPad on the bench and the phone in your pocket — all of them unlock. Buy on vtune-app.com, or directly inside the iOS app, whichever suits.

Signing in

Enter your email on the unlock screen and V-Tune sends you a sign-in link and a 6-digit code. Tap the link on a desktop or web browser; inside the phone and tablet apps, type the code — that's the path that brings you back into the app itself. There is no password to invent or forget.

Already bought it on this device and just reinstalled? Restore purchase, next to Sign in, asks the store directly.

While the trial runs

Nothing nags you until the last three days, when a slim purple countdown appears above the strobe — dismissible, and back on the next launch. To buy before the trial is out, open Settings → V-Tune Pro → Unlock now. Once the trial ends the unlock screen stays up until you buy, sign in or restore.

The onboarding tour holds off while the app is locked, so you never get walked through an app you can't yet use.

14. Tips & troubleshooting

Things to try if something feels off

“The bars are jittery / never lock.”

Background noise is usually the culprit. Open Settings with the gear, then (a) lower Microphone Sensitivity until just your strikes register, and (b) set Hum to your local mains frequency (50 Hz UK/EU, 60 Hz US).

“The cents number is right but the strobe disagrees.”

They're measuring different things. The cents number is a median-filtered, EMA-smoothed reading designed to be a stable digit you can read. The strobe is raw phase rate — it shows instantaneous motion. Trust the strobe for fine adjustments; trust the cents number for the overall verdict.

“It's telling me I'm in tune but I'm clearly not.”

Check Reference A4. If it's set to something exotic (442, 432) you'll be tuning against a different concert pitch — it's in the menu on desktop, or Settings → Tuning on mobile. Also check you're on the right note: handpans have rich overtones, and it's easy to accidentally lock onto a partial that isn't the fundamental. The live note indicator on the note grid (cyan glow) helps catch this.

“The strobe is too lively / too sluggish.”

Settings → Speed is your friend. Drop it for a calmer reading, raise it for a more responsive one.

“I want to tune a partial that isn’t 1x, 2x, or 3x.”

Turn on the Spectrum Analyser, find the peak you care about, and shift+drag (or touch-hold drag on mobile) a bracket around it. V-Tune will generate a dedicated strobe band for that partial.

“I bought V-Tune but this device still shows the trial.”

The unlock travels with your account, so the device needs to know who you are. Open the unlock screen — Settings → V-Tune Pro during the trial — hit Sign in, and use the same email you bought with. If you bought inside the iOS app on this very device, Restore purchase is the quicker route.

“My phone screen has a notch / home indicator.”

Handled — V-Tune respects iOS safe-area insets on the header, the quick-pick slide-up, and the menu. The strobe never hides behind the notch or the home-indicator strip.

That’s the whole app. Strike a note, watch it lock in.